Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University

Purposefully different,

consistently excellent

Welcome to EEN1037: Web Applications Development

HTML & HTML5
Hyper Text Markup Language

Week 2

Joe Mullally

DCU
Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University

HTML (seen in Week 1)

- HyperText Markup Language

```
<!DOCTYPE html>
<html>
        <body>
            <h1>My First Heading</h1>
            <p>My first paragraph.</p>
        </body>
</html>
```

DCU
Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University

Basics (seen in Week 1)

```
<!DOCTYPE html>
<html>
     <head>
          <title>My first page</title>
     </head>
     <body>

          Hello!

     </body>
</html>
```

DCU
Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University

HTML5

- Standardised in October 2014
- Previous standard release (HTML4) was in 1997! (with a revision 4.01 in 2000)
- For the web you would expect faster specification standardisation
- Many big players in the web game – ALOT of money!
- Many browser companies start early implementation of features which are likely to make it into final specifications... Eg. Canvas element usable in Chrome, Safari, Opera and Firefox for a number of years

DCU
Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University

HTML5 Features/Additions

- Markup – range of new elements: <video>, <audio>, <footer>, <nav> <canvas> etc. Others removed such as <font>,
- Drawing on the page: <canvas> - See examples
- Video and Audio: HTML5 browsers to come with built in media player (possible to style using CSS) to play agreed formats
- Application Front-end: Any number of front ends possible through <canvas> element. However, new standard form controls added too
- Offline Storage Database: similar to cookies but with far more data. Enables offline web applications
- <datagrid> for tabular data but allowing sorting, row selection etc.
- Drag and drop – elements on pages or files from desktop/harddrive
- Inline document editing
- Geo-location
- ... Many other features

DCU
Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University

HTML5

- The new standard for HTML
- New features of HTML5 are based on HTML, CSS, DOM, and Javascript
- Goals:
  - Reduce the need for external plugins
  - Provide needed functionalities (upgrade)
  - Coherent across browsers

DCU
Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University

HTML 5

https://www.w3schools.com/html/default.asp

DCU
Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University

HTML5 new tags: Media

- Media elements
  - `<audio>`
  - `<video>`
  - `<source>`
  - `<embed>`
  - `<track>`

DCU
Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University

<video/>

Two important rules

- Use MP4 *always* as the first option to support all iOS devices
- DON’T omit the type attribute, the browser loads the meta data of each source if you do

DCU
Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University

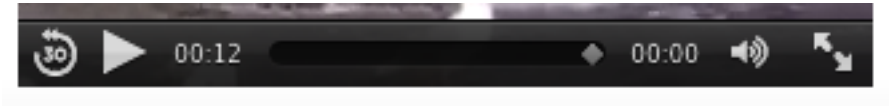
<video> example

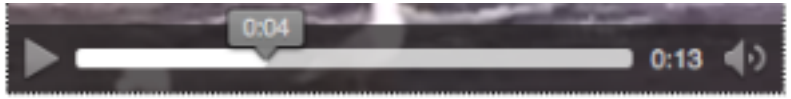
```
<video>
  <source
     src="movie.mp4"
     type="video/mp4" />
  <source
          src="movie.webm”
          type=“video/webm” />
</video>
```

http://www.html5rocks.com/en/tutorials/video/basics/

DCU
Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University

<video/> in browsers





DCU
Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University

Drawing

- <canvas>

DCU
Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University

Canvas example

```
<!DOCTYPE html>
<html>
        <body>
                <canvas id="myCanvas" width="200" height="100”
                         style="border:1px solid #c3c3c3;” />
                <script>
                        var c=document.getElementById("myCanvas");
                        var ctx=c.getContext("2d");
                        ctx.fillStyle="#FF0000";
                        ctx.fillRect(0,0,150,75);
                </script>
        </body>
</html>
```

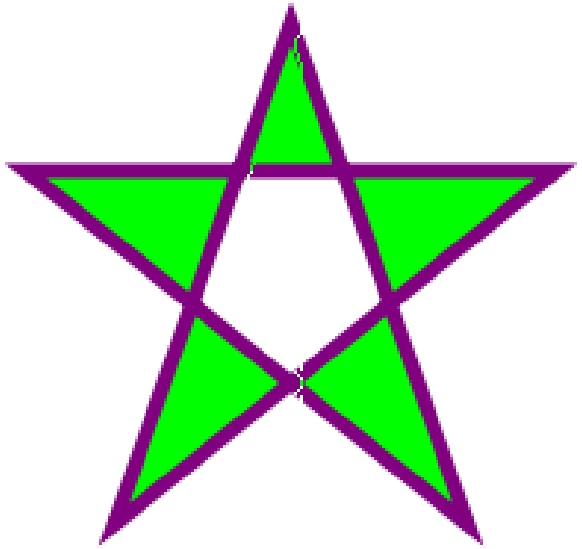
DCU
Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University

Drawing

Scalable

Vector

Graphics



```
<svg xmlns="http://www.w3.org/2000/svg" version="1.1"
height="190">
      <polygon points="100,10 40,180 190,60 10,60 160,180"
             style="fill:lime;stroke:purple;stroke-
width:5;fill-rule:evenodd;">
</svg>
```

DCU
Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University

New input (type) elements

- Color Your favorite color: `<input type=”color” name=”favcolor”>`
- Date, datetime, datetime-local

  Your birthday: `<input type="date" name="bday">` 2012-11-13T08:13 Submit
- Email E-mail: `<input type="email" name="useremail">`
- and more…(see w3schools.com)

DCU
Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University

Structure elements

- <section/>
- <header/>
- <footer/>
- <nav/>
- <article/>

http://www.techrepublic.com/blog/webmaster/html5-using-structural-elements-for-header-footer-and-navigation/549

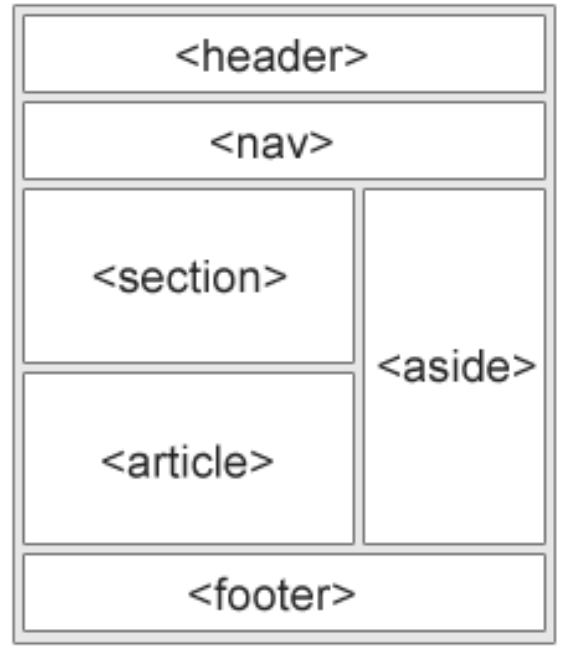
DCU
Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University

New Semantic Elements in HTML5

Many web sites contain HTML code like:
to indicate navigation, header, and footer.

HTML5 offers new semantic elements to define different parts of a web page:

- <article>
- <aside>
- <details>
- <figcaption>
- <figure>
- <footer>
- <header>
- <main>
- <mark>
- <nav>
- <section>
- <summary>
- <time>



DCU
Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University

<article/> and <header/>

```
<article>
  <header>
    <hgroup>
      <h1>Raising Roses</h1>
      <h2>The guide to a perfect rose
garden</h2>
    </hgroup>
 </header>
</article>
```

DCU
Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University

<nav/> example

```
<nav>
 <ul>
   <li><a href="#">Home</a></li>
   <li><a
href="#/events">Events</a></li>
   <li><a href="#/Blog">Blog</a></li>
   <li><a
href="#/Products">Products</a></li>
 </ul>
</nav>
```

DCU
Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University

HTML5 Web Storage

- web pages can store data locally within the user's browser.
- Web Storage is more secure and faster than cookies.
- The data is not included with every server request, but used ONLY when asked for.
- It is also possible to store large amounts of data, without affecting the website's performance.
- The data is stored in key/value pairs, and a web page can only access data stored by itself

DCU
Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University

Some removed elements

- <applet>
-
- <font>
- <frame>, <frameset>, <noframes>
- <strike>

DCU
Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University

Useful links

- Demos: http://html5demos.com/
- W3schools resources: https://www.w3schools.com/html/default.asp
- Graphs: http://www.rgraph.net/examples/index.html
- HTML5 spec: http://dev.w3.org/html5/spec/single-page.html

DCU
Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University